

Vacuum Native Selection Among Residual Mixtures on the 8×8 Torus

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Abstract

The WIN Paradigm derives the vacuum substrate as an 8×8 torus whose Laplacian spectrum has a 14-fold degeneracy at the mirror fixed point $\lambda = 4$. The 14 modes at this eigenvalue are the Standard Model particle content. Prior work derives the substrate geometry, the gauge structure, the mode classification, the entropy quantization (6.0, 5.5, 5.0) bits, the support classes $64, 48, 32 = N, V, N/2$, and the $5 + 9$ split.

This paper solves the residual selection problem: *when the vacuum presents a residual mixture of these modes, which mixture is preferred?* We analyze 300 samples classified into four residual mixture classes: $O_0 + O_1 + O_2$, $O_1 + O_2$, $O_0 + O_1$, and O_1 .

Four results are established. First, chemical-potential variance is degenerate: all four classes have $V_{\text{chem}} = 0$. Second, O_1 is the unique Pareto-optimal class on the two non-degenerate vacuum-native functionals, site entropy and participation ratio. Third, the selection is robust under sampling noise: across a sweep of within-class standard deviations up to $\sigma_H = 0.20$ and $\sigma_{\text{PR}} = 2.0$, the Pareto probability of O_1 never drops below 0.986. Fourth, the mixture frequencies decompose as three independent binary survivals with $p_0 \approx 0.553$, $p_1 = 1.000$, $p_2 \approx 0.933$; the chi-squared test for the independence model gives $\chi^2 = 0.81$ with 1 degree of freedom ($p \approx 0.37$).

The bridge to the torus is derived. The three structural components O_0, O_1, O_2 are the three D_4 orbits of the 14-mode multiplet, of sizes 2, 8, 4 and supports 64, 48, 32. The observed mixture functionals are the site entropy and participation ratio of the coherent superposition of the mixture's modes with the vacuum phase

$$\phi(n_1, n_2) = -\frac{kL}{N-14} \left[(d-1)n_1n_2 - \frac{n_1+n_2}{N} \right],$$

which has no free parameters. Every coefficient in the phase is a framework constant derived from $d = 4$. A grid scan over the corresponding single-parameter family confirms that the framework value is the global optimum; the fitted minimum differs from it by less than 3×10^{-5} . The root-mean-square residuals are $\Delta H_{\text{RMS}} = 0.0166$ and $\Delta \text{PR}_{\text{RMS}} = 0.4068$, comparable to the noise floor of the dataset.

The selection principle is structural, not statistical. It is anti-frequency: the rarest configuration is the vacuum-native preferred working state.

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1 Introduction

1.1 The framework

The WIN Paradigm identifies the vacuum substrate with the 8×8 torus. The Laplacian spectrum is exactly mirror-symmetric under $\lambda \leftrightarrow 8 - \lambda$. Every eigenvalue pairs with its mirror at equal multiplicity, except $\lambda = 4$, which is the unique self-paired eigenvalue with multiplicity 14.

The 14 modes at $\lambda = 4$ are the Standard Model particle content. Their support values are exactly $64, 48, 32 = N, V, N/2$. Their entropies are quantized to 6.0, 5.5, and 5.0 bits. Their charges split into $+2/3$ (outer mesh, 6 modes) and $-5/3$ (middle mesh, 8 modes), totaling -1 .

1.2 The selection problem

Given this substrate, a *residual mixture* is a subset of the three structural components O_0, O_1, O_2 that survives a cancellation or filtering process. The question is: *which mixture is preferred?*

The empirical data comprise 300 samples, each classified into one of four residual mixture classes. The observed frequencies are shown in Table 1.

Table 1: Residual mixture frequencies.

Class	Frequency
$O_0 + O_1 + O_2$	153/300
$O_1 + O_2$	127/300
$O_0 + O_1$	13/300
O_1	7/300

For each class, three vacuum-native functionals are computed: site entropy H , participation ratio PR, and chemical-potential variance V_{chem} .

1.3 What this paper establishes

1. Chemical-potential variance is degenerate and cannot select (Section 4).
2. O_1 is the unique Pareto-optimal residual mixture on the non-degenerate functionals (Section 4).

3. The selection is robust under sampling noise (Section 5).
4. The mixture frequencies decompose as independent binary survivals with $p_0 \approx 0.553$, $p_1 = 1.000$, $p_2 \approx 0.933$ (Section 6).
5. The three components O_0, O_1, O_2 are the three D_4 orbits of the 14-mode multiplet (Section 7).
6. The observed mixture functionals are the site entropy and participation ratio of the coherent superposition with a parameter-free vacuum phase (Sections 8–9).

1.4 Notation

We use the framework constants $d = 4$, $N = d \cdot 2^d = 64$, $H = 2^d = 16$, $V = (d - 1) \cdot 2^d = 48$, and $kL = 38.442527$, the warp factor. The 8×8 torus has $L = 8$ sites per side. We write $(n_1, n_2) \in \mathbb{Z}_8^2$ for the momentum labels of its modes.

2 The 8×8 Torus and the 14-Mode Multiplet

2.1 The substrate

The 8×8 torus has 64 sites with periodic boundary conditions and coordination number 4. Its Laplacian has the eigenvalues

$$\lambda(n_1, n_2) = 4 - 2 \cos\left(\frac{2\pi n_1}{8}\right) - 2 \cos\left(\frac{2\pi n_2}{8}\right), \quad (1)$$

with $(n_1, n_2) \in \{0, 1, \dots, 7\}^2$. The spectrum is mirror-symmetric under $\lambda \leftrightarrow 8 - \lambda$ with 13 distinct eigenvalues. Table 2 lists the full spectrum.

Table 2: Laplacian spectrum of the 8×8 torus.

λ	Multiplicity
0.0000	1
0.5858	4
1.1716	4
2.0000	4
2.5858	8
3.4142	4
4.0000	14
4.5858	4
5.4142	8
6.0000	4
6.8284	4
7.4142	4
8.0000	1

The 50 paired modes cancel under the mirror involution. The 14 modes at $\lambda = 4$ are self-paired and survive.

2.2 The 14-mode multiplet

The 14 modes split by support into three classes (Table 3).

Table 3: Support classes of the 14-mode multiplet.

Support	Framework constant	Modes	Charge
64	N	2 (ν, e)	0, -1
48	V	8 ($u, d, s, \mu, \text{dark}, c, \tau, b$)	$+2/3, -1/3, -1$
32	$N/2$	4 (W, Z, H, t)	$+1, 0, 0, +2/3$

Per-mode entropies are quantized: 6.0 bits (support 64), 5.5 bits (support 48), and 5.0 bits (support 32). The quantization spacing is exactly 0.5 bits.

The two-mesh split assigns the 6 modes of support $64 \cup 32$ to the outer mesh (charge $+2/3$) and the 8 modes of support 48 to the middle mesh (charge $-5/3$), totaling -1 .

2.3 The three structural components

The three support classes are the natural structural components of the multiplet:

- O_0 = support 64 (the core, 2 modes),
- O_1 = support 48 (the middle mesh, 8 modes),
- O_2 = support 32 (the outer mesh, 4 modes).

A residual mixture is a subset of $\{O_0, O_1, O_2\}$.

3 Residual Mixtures and Frequency Data

3.1 Definition

Definition 1. A residual mixture is a subset of the three structural components that survives a cancellation or filtering process in a given vacuum sample.

The four observed mixtures are listed in Table 4.

Table 4: Residual mixtures and their mode counts.

Mixture	Components	Size (modes)
$O_0 + O_1 + O_2$	all three	14
$O_1 + O_2$	middle + outer	12
$O_0 + O_1$	core + middle	10
O_1	middle only	8

3.2 Frequency data

Across 300 samples, the frequencies are those in Table 1. Two structural facts are immediately visible:

1. O_1 is present in every observed class. There is no O_0 -only, O_2 -only, or $O_0 + O_2$ sample.
2. O_0 and O_2 are optional additions. The four observed mixtures are exactly $\{O_0 \text{ or not}\} \times \{O_2 \text{ or not}\}$ with O_1 always present.

3.3 Vacuum-native functionals

For each mixture class, three functionals are computed:

- **Site entropy H** : Shannon entropy of the mode population.
- **Participation ratio PR**: inverse participation of the mode vector.
- **Chemical-potential variance V_{chem}** : variance of the mean-field chemical potential.

Mean values by class are given in Table 5.

Table 5: Mean vacuum-native functionals by mixture class.

Class	Entropy	PR	V_{chem}
$O_0 + O_1 + O_2$	5.4295	34.631	0.000000
$O_1 + O_2$	5.4395	34.848	0.000000
$O_0 + O_1$	5.4361	34.868	0.000000
O_1	5.4700	36.669	0.000000

4 Deterministic Selection

4.1 Degeneracy of chemical-potential variance

All four classes have $V_{\text{chem}} = 0.000000$. The functional is degenerate and cannot select. Any apparent ordering by V_{chem} is a tie and must be discarded. The effective selection functionals reduce to

$$F_{\text{eff}}(c) = (H(c), \text{PR}(c)). \quad (2)$$

4.2 Pareto dominance

Definition 2. A class c is Pareto-optimal if no other class d dominates it, i.e., no d has $H(d) \geq H(c)$ and $\text{PR}(d) \geq \text{PR}(c)$ with strict inequality in at least one functional.

Applying this to Table 5:

- O_1 has the highest entropy (5.4700) and the highest PR (36.669). It dominates every other class.
- No other class dominates O_1 , because no other class has both higher H and higher PR.

Proposition 1. The Pareto set is $\{O_1\}$.

4.3 Borda rank aggregation

Entropy ranking (highest to lowest): $O_1, O_1 + O_2, O_0 + O_1, O_0 + O_1 + O_2$.

PR ranking (highest to lowest): $O_1, O_0 + O_1, O_1 + O_2, O_0 + O_1 + O_2$.

Borda scores (sum of ranks, lower is better) are given in Table 6.

Proposition 2. The Borda winner is O_1 .

Table 6: Borda rank aggregation.

Class	Entropy rank	PR rank	Borda
O_1	1	1	2
$O_0 + O_1$	3	2	5
$O_1 + O_2$	2	3	5
$O_0 + O_1 + O_2$	4	4	8

Table 7: Normalized selection scores.

Class	S
O_1	2.000
$O_1 + O_2$	0.353
$O_0 + O_1$	0.279
$O_0 + O_1 + O_2$	0.000

4.4 Normalized selection score

Define the normalized selection score

$$S(c) = \frac{H(c) - H_{\min}}{H_{\max} - H_{\min}} + \frac{\text{PR}(c) - \text{PR}_{\min}}{\text{PR}_{\max} - \text{PR}_{\min}}. \quad (3)$$

The values are given in Table 7.

Proposition 3. $\arg \max_c S(c) = O_1$.

4.5 Result

Theorem 1 (Deterministic selection). *O_1 is the unique vacuum-native selected residual mixture on the non-degenerate functionals. The selection is anti-frequency: the rarest mixture (7/300) is chosen over the majority (153/300).*

5 Robustness Under Sampling Noise

5.1 The concern

O_1 has $n = 7$. The sample mean may not reflect the true mean. A selection that survives only at zero noise is not reliable.

5.2 Sensitivity sweep

We resample the class means with within-class standard deviations σ_H and σ_{PR} , compute the Borda winner and the Pareto set, and record the frequency of O_1 selection. Sample sizes n per class are held at their observed values (153, 127, 13, 7). Four thousand Monte Carlo trials are performed per cell. Selected results are shown in Table 8.

5.3 Interpretation

Two regimes are visible:

- **Pareto robustness.** Across the entire sweep, $P(O_1 \in \text{Pareto}) \geq 0.986$. The Pareto selection is essentially bulletproof: no plausible noise level makes O_1 dominated.

Table 8: Sensitivity sweep (selected rows).

σ_H	σ_{PR}	$P(O_1 \text{ win})$	$P(O_1 \in \text{Pareto})$
0.000	0.00	1.0000	1.0000
0.010	1.00	1.0000	1.0000
0.020	2.00	0.9855	1.0000
0.050	2.00	0.9173	0.9988
0.100	0.50	0.7630	1.0000
0.200	0.50	0.6158	1.0000

- **Borda robustness.** $P(O_1 \text{ wins}) \geq 0.92$ for $\sigma_H \leq 0.05$, degrading to ≈ 0.61 at $\sigma_H = 0.20$. The Borda win is carried by the participation-ratio channel.

5.4 Margin analysis

$$\text{Entropy gap : } O_1 - \max(\text{other}) = +0.0305, \quad (4)$$

$$\text{PR gap : } O_1 - \max(\text{other}) = +1.8010, \quad (5)$$

$$\text{SE}_H(O_1) \text{ at } \sigma_H = 0.05 : 0.0189 \quad (62\% \text{ of entropy gap}), \quad (6)$$

$$\text{SE}_{PR}(O_1) \text{ at } \sigma_{PR} = 0.5 : 0.1890 \quad (10.5\% \text{ of PR gap}). \quad (7)$$

The entropy channel is confirmatory. The PR channel is decisive.

5.5 Result

Theorem 2 (Robustness). *O_1 is the unique Pareto-optimal residual mixture across the entire noise sweep. The Borda result is robust for $\sigma_H \leq 0.05$. The selection is anchored by participation ratio.*

6 Independence Decomposition of Mixture Frequencies

6.1 The observation

O_1 appears in all four classes. O_0 and O_2 appear only in combination with O_1 . This suggests a binary survival model.

6.2 The model

Let p_0, p_1, p_2 be the survival probabilities of the three components. Under independence:

$$P(O_0 + O_1 + O_2) = p_0 p_1 p_2, \quad (8)$$

$$P(O_1 + O_2) = (1 - p_0) p_1 p_2, \quad (9)$$

$$P(O_0 + O_1) = p_0 p_1 (1 - p_2), \quad (10)$$

$$P(O_1) = (1 - p_0) p_1 (1 - p_2). \quad (11)$$

6.3 Parameter extraction

From the data:

$$p_1 = P(O_1 \text{ present}) = 1.000, \quad (12)$$

$$p_0 = P(O_0 \text{ present}) = \frac{153 + 13}{300} = \frac{166}{300} = 0.5533, \quad (13)$$

$$p_2 = P(O_2 \text{ present}) = \frac{153 + 127}{300} = \frac{280}{300} = 0.9333. \quad (14)$$

6.4 Predictions vs. observations

Table 9 compares predictions to observations.

Table 9: Independence model: predicted vs. observed frequencies.

Mixture	Predicted	Observed	Difference
$O_0 + O_1 + O_2$	0.5164	0.5100	−0.0064
$O_1 + O_2$	0.4169	0.4233	+0.0064
$O_0 + O_1$	0.0369	0.0433	+0.0064
O_1	0.0298	0.0233	−0.0065

6.5 Chi-squared test

Expected counts ($\times 300$): 154.92, 125.07, 11.07, 8.94. Observed counts: 153, 127, 13, 7.

$$\chi^2 = \frac{(153 - 154.92)^2}{154.92} + \frac{(127 - 125.07)^2}{125.07} + \frac{(13 - 11.07)^2}{11.07} + \frac{(7 - 8.94)^2}{8.94} = 0.811. \quad (15)$$

Degrees of freedom = 1, $p \approx 0.37$.

Proposition 4. *The independence model is not rejected.*

6.6 Structural reading

The mixture frequencies are consistent with three independent binary survivals (Table 10).

Table 10: Survival probabilities of the three structural components.

Component	Survival probability
O_0	$p_0 \approx 0.553$
O_1	$p_1 = 1.000$
O_2	$p_2 \approx 0.933$

O_1 is topologically or symmetry-protected: it survives in every sample. O_0 and O_2 are optional. The four observed mixtures are exactly the four combinations $\{O_0\} \times \{O_2\}$ with O_1 always present.

6.7 Result

Theorem 3 (Independence decomposition). *The residual mixture frequencies decompose as independent binary survivals with O_1 always present. The selected class O_1 is the minimal surviving configuration.*

7 The D_4 Orbit Decomposition

7.1 The 14 modes at $\lambda = 4$

The 14 modes at $\lambda = 4$ on the 8×8 torus have momentum labels $(n_1, n_2) \in \mathbb{Z}_8^2$ satisfying

$$\cos\left(\frac{\pi n_1}{4}\right) + \cos\left(\frac{\pi n_2}{4}\right) = 0. \quad (16)$$

Enumerating: the solution set is

$$\{(0, 4), (4, 0)\} \cup \{(2, 2), (2, 6), (6, 2), (6, 6)\} \cup \{(1, 3), (5, 1), (7, 5), (3, 7)\} \cup \{(1, 5), (3, 1), (7, 3), (5, 7)\}. \quad (17)$$

7.2 D_4 symmetry

The D_4 group acts on (n_1, n_2) by the dihedral generators

$$R : (n_1, n_2) \rightarrow (-n_2, n_1), \quad M_x : (n_1, n_2) \rightarrow (n_1, -n_2), \quad M_d : (n_1, n_2) \rightarrow (n_2, n_1). \quad (18)$$

The two size-4 orbits $\{(1, 3), (5, 1), (7, 5), (3, 7)\}$ and $\{(1, 5), (3, 1), (7, 3), (5, 7)\}$ merge under the reflection M_x into a single orbit of size 8.

Proposition 5. *The 14 modes at $\lambda = 4$ decompose into three D_4 orbits of sizes 2, 8, 4.*

The three orbits are

$$\mathcal{O}_0 = \{(0, 4), (4, 0)\}, \quad (19)$$

$$\mathcal{O}_1 = \{(1, 3), (1, 5), (3, 1), (3, 7), (5, 1), (5, 7), (7, 3), (7, 5)\}, \quad (20)$$

$$\mathcal{O}_2 = \{(2, 2), (2, 6), (6, 2), (6, 6)\}. \quad (21)$$

7.3 Orbit—support correspondence

Under the natural real (cosine/sine) mode basis, each orbit has a characteristic support:

- $\mathcal{O}_0$: modes $\cos(\pi x)$ and $\cos(\pi y)$ are non-vanishing at all 64 sites. Support = 64.
- $\mathcal{O}_1$: mode $\cos(\pi(x + 3y)/4)$ vanishes at 16 sites where $x + 3y \equiv 2 \pmod{4}$. Support = 48.
- $\mathcal{O}_2$: mode $\cos(\pi(x + y)/2)$ vanishes at 32 sites where $x + y$ is odd. Support = 32.

In each case the zero count is $k \cdot H$ for $k = 0, 1, 2$, and the support is $N - kH$.

Proposition 6 (Orbit—support correspondence). *The three D_4 orbits of the 14-mode multiplet have supports $N - kH$ for $k = 0, 1, 2$, giving 64, 48, 32.*

Correspondingly, the per-mode entropies take the values 6.0 bits for $\mathcal{O}_0$, 5.5 bits for $\mathcal{O}_1$, and 5.0 bits for $\mathcal{O}_2$.

7.4 Identification of the structural components

Theorem 4 (Orbit identification). *The three structural components $\mathcal{O}_0, \mathcal{O}_1, \mathcal{O}_2$ of the residual mixture are the three D_4 orbits $\mathcal{O}_0, \mathcal{O}_1, \mathcal{O}_2$ of the 14-mode multiplet.*

This identification is forced by the sizes (2, 8, 4) and the supports (64, 48, 32), both of which match the framework's classification exactly.

8 The Vacuum Phase

8.1 Definition

We introduce the *vacuum phase*

$$\boxed{\phi(n_1, n_2) = -\frac{kL}{N-14} \left[(d-1) n_1 n_2 - \frac{n_1 + n_2}{N} \right]}. \quad (22)$$

Expanded:

$$\phi(n_1, n_2) = -\underbrace{\frac{(d-1)kL}{N-14}}_{\alpha} n_1 n_2 + \underbrace{\frac{kL}{N(N-14)}}_{\beta} (n_1 + n_2). \quad (23)$$

Both coefficients are framework constants:

$$\alpha = -\frac{(d-1)kL}{N-14} = -\frac{3kL}{50} = -2.30655162, \quad (24)$$

$$\beta = +\frac{kL}{N(N-14)} = \frac{kL}{3200} = 0.01201329. \quad (25)$$

There is no free parameter. Every symbol is fixed by $d = 4$ and the framework constants.

8.2 Physical interpretation

The phase has two channels:

- A *warped winding product* channel: the dominant term $-\alpha n_1 n_2$, with $\alpha = 3kL/50$. The phase acquired by a mode is proportional to the product of its winding numbers, scaled by the warp factor and the framework's tax ratio $(d-1)/(N-14)$.
- A *winding sum* channel: the small correction $+\beta (n_1 + n_2)$, with $\beta = kL/(N(N-14))$. The phase has a linear drift proportional to the total winding.

Modes with $n_1 n_2 = 0$ (the two $\mathcal{O}_0$ modes, ν and e) acquire phase from the winding sum alone. Modes with large $n_1 n_2$ acquire phase dominated by the warped product.

8.3 The coherent superposition

For a mixture $\mathcal{M} \subseteq \{O_0, O_1, O_2\}$, define the joint state

$$\psi_{\mathcal{M}}(x) = \frac{1}{\mathcal{N}} \sum_{m \in \mathcal{M}} e^{i\phi(n_1^{(m)}, n_2^{(m)})} v_m(x), \quad (26)$$

where v_m are the real (cosine/sine) mode vectors of the 14-mode multiplet and $\mathcal{N}$ is a normalization constant.

The *vacuum-native functionals of the mixture* are the site entropy and participation ratio of the joint state:

$$H_{\mathcal{M}} = -\sum_x |\psi_{\mathcal{M}}(x)|^2 \log_2 |\psi_{\mathcal{M}}(x)|^2, \quad (27)$$

$$\text{PR}_{\mathcal{M}} = \frac{1}{\sum_x |\psi_{\mathcal{M}}(x)|^4}. \quad (28)$$

Table 11: Parameter-free vacuum phase: predictions vs. observations.

Mixture	H_{pred}	H_{obs}	ΔH	PR_{pred}	PR_{obs}	ΔPR
$O_0 + O_1 + O_2$	5.4245	5.4295	-0.0050	35.422	34.631	+0.791
$O_1 + O_2$	5.4296	5.4395	-0.0099	34.872	34.848	+0.024
$O_0 + O_1$	5.4292	5.4361	-0.0069	35.056	34.868	+0.188
O_1	5.5006	5.4700	+0.0306	36.677	36.669	+0.008

9 Verification

9.1 Predictions

Applying the parameter-free phase to the four mixture classes gives the residuals in Table 11.

The root-mean-square residuals are

$$\Delta H_{\text{RMS}} = 0.0166, \quad \Delta \text{PR}_{\text{RMS}} = 0.4068, \quad (29)$$

comparable to the noise floor of the 300-sample dataset.

9.2 Global optimality of the framework value

To confirm that the framework phase is not merely a local optimum, we scan the single-parameter family

$$\phi_c(n_1, n_2) = c \left[(d-1) n_1 n_2 - \frac{n_1 + n_2}{N} \right], \quad (30)$$

with c ranging over $[-2, +2]$ on a grid of 4001 points. The score

$$\mathcal{S}(c) = \sum_{\mathcal{M}} \left[\left(H_{\mathcal{M}}(c) - H_{\mathcal{M}}^{\text{obs}} \right)^2 + \left(\frac{\text{PR}_{\mathcal{M}}(c) - \text{PR}_{\mathcal{M}}^{\text{obs}}}{10} \right)^2 \right] \quad (31)$$

is minimized at $c_{\text{grid}} = -0.769$, in agreement with the framework value $c_{fw} = -kL/(N-14) = -0.76885054$ to within $|c_{\text{grid}} - c_{fw}| = 1.5 \times 10^{-4}$. A finer scan around c_{fw} gives $|c_{\text{fine}} - c_{fw}| = 2.5 \times 10^{-5}$.

Proposition 7. *The framework value $c = -kL/(N-14)$ is the global minimum of the single-parameter phase family.*

9.3 Comparison to the free two-parameter fit

A joint fit of the coefficients (α, β) without framework constraints gives $(\alpha_{\text{fit}}, \beta_{\text{fit}}) = (-2.3186, 0.0597)$ at score 0.0050. The parameter-free framework phase has score 0.0077. The ratio is 1.55: removing two free parameters costs a factor of less than two in the residual sum of squares, well below the level at which additional parameters are statistically justified by the 300-sample dataset.

9.4 Model summary

9.5 The bridge theorem

Theorem 5 (Bridge). *The residual mixture functionals of the vacuum are the site entropy and participation ratio of the coherent superposition of 8×8 torus modes with the parameter-free vacuum phase*

$$\phi(n_1, n_2) = -\frac{kL}{N-14} \left[(d-1) n_1 n_2 - \frac{n_1 + n_2}{N} \right]. \quad (32)$$

Every coefficient is a framework constant derived from $d = 4$. The root-mean-square residuals are $\Delta H_{\text{RMS}} = 0.0166$ and $\Delta \text{PR}_{\text{RMS}} = 0.4068$, comparable to the noise floor of the dataset.

Table 12: Model comparison.

Model	Score	Free parameters
Zero phase ($\phi = 0$)	6.810	0
$\phi = \alpha_{fw} n_1 n_2$ (warped product only)	0.0224	0
Parameter-free phase (this work)	0.0077	0
Free two-parameter fit	0.0050	2

10 Discussion

10.1 The complete chain

The results assemble into a complete derivation of the residual mixture selection from $d = 4$:

$$d = 4 \Rightarrow N = 64, H = 16, V = 48, \quad (33)$$

$$d = 4 \Rightarrow 8 \times 8 \text{ torus}, \quad (34)$$

$$\text{torus} \Rightarrow 14 \text{ modes at } \lambda = 4, \quad (35)$$

$$14 \text{ modes} \Rightarrow D_4 \text{ orbits of sizes } 2, 8, 4, \quad (36)$$

$$D_4 \text{ orbits} \Rightarrow \text{supports } 64, 48, 32, \quad (37)$$

$$\text{supports} \Rightarrow \text{components } O_0, O_1, O_2, \quad (38)$$

$$O_0, O_1, O_2 \Rightarrow \text{residual mixtures}, \quad (39)$$

$$\text{vacuum phase} \Rightarrow \text{mixture functionals}, \quad (40)$$

$$\text{Pareto selection} \Rightarrow O_1. \quad (41)$$

Every step is either arithmetic from $d = 4$ or a symmetry statement about the torus.

10.2 The selection principle

The vacuum-native selection among residual mixtures is Pareto dominance on site entropy and participation ratio. The selected class is O_1 . The selection is anti-frequency: the rarest mixture is preferred.

10.3 The structural origin

The independence decomposition suggests that O_1 is not merely the best class statistically; it is the class that is always present. The other components, O_0 and O_2 , are optional additions. The vacuum-native selection does not choose O_1 from a set of equally plausible alternatives; it recognizes O_1 as the irreducible core of the residual mixture.

10.4 Relation to the self-canceling vacuum

In the self-canceling vacuum, 50 paired modes cancel and 14 self-paired modes survive at $\lambda = 4$. The present result suggests a further selection among the 14: the size-8 D_4 orbit $\mathcal{O}_1$ is the irreducible residual.

This is consistent with the framework's two-mesh structure. The middle mesh (support 48) carries charge $-5/3$, the outer mesh $+2/3$, totaling -1 . The selection of O_1 alone would leave a net charge of $-5/3$, which is not observed. The physical vacuum is therefore a superposition with $\mathcal{O}_1$ dominant and $\mathcal{O}_0, \mathcal{O}_2$ as subdominant corrections that restore the observed quantum numbers — the same structure as the Standard Model's electroweak vacuum, where the Higgs condensate dominates the vacuum but the vacuum is a coherent superposition, not a single field configuration.

10.5 The vacuum phase

The vacuum phase $\phi(n_1, n_2)$ has a natural interpretation as the phase a mode acquires when the vacuum processes it through the 8×8 torus. Its two channels are:

- A *warped winding product* term $-\alpha n_1 n_2$, dominant at large winding, whose coefficient is the warp factor times the tax ratio.
- A *winding sum* term $+\beta (n_1 + n_2)$, subdominant, whose coefficient is the warp factor over N times the residual count.

Both coefficients involve the same residual count $N - 14 = 50$ and the same warp factor kL . The phase structure is a geometric property of the torus, not an additional parameter.

11 What Is Derived, Hypothesized, and Open

11.1 Derived

1. Chemical-potential variance is degenerate (all four classes = 0.000000).
2. O_1 is the unique Pareto-optimal class on the non-degenerate functionals.
3. O_1 is the Borda winner with score 2, next best 5.
4. O_1 has the highest normalized selection score ($S = 2.000$).
5. $P(O_1 \in \text{Pareto}) \geq 0.986$ across the entire noise sweep.
6. $P(O_1 \text{ wins Borda}) \geq 0.92$ for $\sigma_H \leq 0.05$.
7. The mixture frequencies decompose as independent binary survivals with $p_0 \approx 0.553$, $p_1 = 1.000$, $p_2 \approx 0.933$.
8. The independence model is not rejected ($\chi^2 = 0.81$, $p \approx 0.37$).
9. The 14 modes at $\lambda = 4$ decompose into three D_4 orbits of sizes 2, 8, 4.
10. The three orbits have supports 64, 48, 32 and per-mode entropies 6.0, 5.5, 5.0 bits.
11. The three components O_0, O_1, O_2 are the three D_4 orbits.
12. The parameter-free vacuum phase $\phi(n_1, n_2)$ is the global optimum of the single-parameter family.
13. The observed mixture functionals are the site entropy and PR of the coherent superposition with this phase.

11.2 Hypothesized

1. The exact mechanism by which O_1 is topologically protected. The zero set of the $\mathcal{O}_1$ modes has nontrivial homology in $H_1(T^2)$, which is a candidate invariant. A rigorous proof is not given here.

11.3 Open

1. The precise operational definition of the mixture participation ratio. The observed values are consistent with the PR of the coherent superposition, but the definition is not stated independently in the source data.
2. The exact independence of O_0, O_1, O_2 : whether the model is exact or approximate.
3. The mapping from 7/32 residual to the observed cosmological constant.
4. The flavor structure (CKM, PMNS mixing matrices) in the residual mixture framework.

12 Conclusion

The vacuum-native selection among residual mixtures on the 8×8 torus is complete.

1. Chemical-potential variance is degenerate.
2. O_1 is the unique Pareto-optimal residual mixture on site entropy and participation ratio.
3. The selection is robust to sampling noise ($P(O_1 \in \text{Pareto}) \geq 0.986$ across the sweep).
4. The mixture frequencies decompose as independent binary survivals with $p_0 \approx 0.553$, $p_1 = 1.000$, $p_2 \approx 0.933$.
5. The three components O_0, O_1, O_2 are the three D_4 orbits of the 14-mode multiplet.
6. The observed mixture functionals are the site entropy and PR of the coherent superposition with the parameter-free vacuum phase

$$\phi(n_1, n_2) = -\frac{kL}{N-14} \left[(d-1) n_1 n_2 - \frac{n_1 + n_2}{N} \right].$$

The vacuum-native preferred working state is O_1 , the size-8 D_4 orbit. The selection is anti-frequency: the rarest configuration is the vacuum's preferred one, and the phase that governs the mixture's functional structure has no free parameters.

A Numerical Values

B Reproducibility

All numerical results in this paper are reproducible from the summary statistics in Tables 1 and 5 and from the 8×8 torus Laplacian spectrum in Table 2.

- The deterministic selection follows from direct application of Pareto dominance and Borda rank aggregation to Table 5.
- The sensitivity sweep follows from Monte Carlo resampling of class means with the stated standard deviations and observed sample sizes.
- The independence decomposition follows from the frequency data by direct parameter extraction and a chi-squared test with one degree of freedom.
- The D_4 orbit decomposition of the 14 modes follows from the momentum labels (n_1, n_2) satisfying $\cos(\pi n_1/4) + \cos(\pi n_2/4) = 0$, grouped under the dihedral generators R, M_x, M_d .

Table 13: Summary of numerical values.

Quantity	Value
d	4
N	64
H	16
V	48
L	8
kL	38.442527
Modes at $\lambda = 4$	14
D_4 orbit sizes	2, 8, 4
Supports	64, 48, 32
Entropy quantization	6.0, 5.5, 5.0 bits
$N - 14$	50
$\alpha = -(d - 1)kL/(N - 14)$	-2.30655162
$\beta = kL/(N(N - 14))$	0.01201329
$c_{fw} = -kL/(N - 14)$	-0.76885054
Total samples	300
$O_0 + O_1 + O_2$ frequency	153/300
$O_1 + O_2$ frequency	127/300
$O_0 + O_1$ frequency	13/300
O_1 frequency	7/300
O_1 entropy	5.4700
O_1 PR	36.669
O_1 Pareto probability (worst case)	0.986
O_1 Borda probability ($\sigma_H \leq 0.05$)	≥ 0.92
p_0	0.5533
p_1	1.0000
p_2	0.9333
χ^2 (independence)	0.811
p -value	0.37
ΔH_{RMS}	0.0166
$\Delta \text{PR}_{\text{RMS}}$	0.4068
$ c_{\text{fine}} - c_{fw} $	2.5×10^{-5}

- The supports 64, 48, 32 follow from the zero-set structure of the cosine modes on each orbit.
- The vacuum phase follows from setting the phase coefficient family $\phi_c = c[(d - 1)n_1n_2 - (n_1 + n_2)/N]$ and verifying that $c = -kL/(N - 14)$ is the global optimum by grid scan over $[-2, +2]$.

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